

Khatam: Reducing the Communication Complexity of Code-Based SNARKs

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Abstract

Two techniques have recently emerged in the construction of Succinct Non-Interactive Arguments of Knowledge (SNARKs) that yield extremely fast provers; The use of multilinear (instead of univariate) polynomial commitment schemes (PCS) and the construction of PCS from error-correcting codes. Recently, BaseFold (Crypto 2024) introduced a family of PCS that combine these two techniques, thereby achieving a better trade-off between prover time and verifier costs than the state of the art. Despite its impressive overall efficiency, BaseFold suffered from larger proof sizes than its univariate counterparts, due to unproven claims about linear codes, which were not relevant in the univariate setting.

This work closes this gap by proving a new fact about linear codes – that if π_L, π_R are two vectors in $\mathbb{F}^n$ and if $\pi_L + r\pi_R$ is close to a codeword in C , then π_L, π_R and $(\pi_L + r\pi_R)$ all agree with codewords at positions in the same set $S \subset [n]$, except with negligible probability over $r \leftarrow \mathbb{F}$. Our result holds as long as $|S| > (1 - \Delta_C + \epsilon)^{1/3} + \eta$, for $\epsilon, \eta \in [0, 1]$ and with failure probability smaller than $\frac{1}{\epsilon\eta|\mathbb{F}|}$, where Δ_C is the minimum distance of the code. Furthermore, our results extend to any finite field and any linear code.

1 Introduction

In recent years, error-correcting codes have emerged as a key ingredient in the construction of efficient, practice-oriented, SNARKs. A prover of a code-based SNARK commits to its witness by encoding it with a linear error-correcting code, which uses relatively cheap operations such as finite-field addition and multiplication. The verifier tests the proximity of the prover’s codeword to the error-correcting code by engaging with the prover in an interactive oracle proof of proximity¹(IOPP) [5, 22]. Then, using a collision-resistant hash function, we obtain a polynomial commitment scheme (PCS) [18], where a prover commits to a polynomial $P \in \mathbb{F}[X]$ so that it can later prove evaluation claims of the form $P(\alpha) = \beta$. Finally, a PCS compiles a polynomial interactive oracle proof (PIOP) into a SNARK. We refer to [7, 11, 24, 25] for more details on this transformation.

Despite their impressive prover efficiency, verifier costs remain a major bottleneck in code-based SNARKs, due mainly to the query complexity of the underlying IOPP. IOPPs have multiple rounds; in each round the prover sends an oracle in response to verifier randomness and the verifier queries and tests elements from these oracles. It would be too expensive for

¹It may be useful to think of an IOPP as a PCPP [4, 13] but with multiple interactive rounds;

the verifier to query each element from the oracle. Instead the verifier obtains a probabilistic guarantee that a large fraction, $\beta \in [0, 1]$, of elements from each prover oracle pass its tests. We choose the number of verifier queries to be l such that $\beta^l < 2^{-\lambda}$, where λ is a security parameter of our choosing. Our goal is to prove security with the smallest possible value of l , which also requires us to use the smallest possible value of β . In the Fast Reed-Solomon IOPP(FRI) [6], β can be as small as $\sqrt{1 - \Delta_C}$, where Δ_C is the minimum distance of the code. This setting is the best proven result and is commonly referred to as the “list-decoding” regime, since by the Johnson Bound, there is only a small *list* of codewords that agree with any vector in more than $\sqrt{1 - \Delta_C}$ fraction of locations. Alternatively, we refer to the case when the lower bound of β is $(1 - \Delta_C/2)$ (which requires more verifier queries and larger proofs) as the “unique-decoding regime”, as there is only one unique codeword that agrees with a vector in that many locations (due to the distance properties of the error-correcting code).

Since the FRI IOPP is proven secure in the *list-decoding regime*, its verifier is both asymptotically and concretely efficient. However, the FRI IOPP can only be directly used² as a univariate PCS, and SNARKs based on univariates have a higher overhead than those based on multilinear PCS, which were introduced by [21]³. BaseFold [25], introduced a technique for using FRI directly as a multilinear PCS by interleaving the sum-check protocol [20] for multilinear polynomial evaluation with (a generalization of) FRI. It avoids the overhead of univariate SNARKs while maintaining polylogarithmic communication. Several works ([2, 12, 17, 23]) have already adapted BaseFold to different settings. However, BaseFold was originally only proven secure in the unique decoding regime and as a result, its proofs were concretely larger than FRI’s. For that matter, no⁴ multilinear polynomial commitment scheme had been proven secure in the list-decoding regime when the first version of this work was first made public. In particular, Brakedown [15], Liger [1], and Blaze [10] are only proven secure in the unique decoding regime.

1.1 Our Contributions

In this work, we prove a stronger version of Lemma 3.2 (One-and-half Johnson distance preservation) from Deep-FRI([8]), which states the following. For a vector $\mathbf{v} \in \mathbb{F}^n$, denote by $\Delta(\mathbf{v}, C)$ the distance from $\mathbf{v}$ to its closest codeword in C , breaking ties arbitrarily. For $\beta \in [0, 1]$, $\pi_L, \pi_R \in \mathbb{F}^n$, if $\Delta(\pi_L + r\pi_R, C) < 1 - \beta$, then except with negligible probability over choice of $r \leftarrow \mathbb{F}$, there exists $c_L, c_R \in C$ such that

$$|\{i \in [n] : \pi_L[i] = c_L[i] \wedge \pi_R[i] = c_R[i]\}| > (\beta - \eta)n,$$

where $\beta, \eta \in [0, 1]$. In this paper, we prove something stronger - that if $(\pi_L + r\pi_R)$ agree with a codeword in C on a set S , then except with negligible probability over choice of $r \leftarrow \mathbb{F}$, then π_L, π_R agree with codewords in C on some large subset of S . In other words, our result guarantee that $\pi_L, \pi_R, \pi_L + r\pi_R$ have a large overlapping agreement set. To state this more formally, we prove that if $(\pi_L + r\pi_R)$ agree with a codeword in C on a set S , then except with negligible probability over choice of $r \leftarrow \mathbb{F}$, that there exists $c_L, c_R \in C$ and $S' \subset S$, with $|S'| > \beta - \eta$ such that

$$\pi_L[S'] = c_L[S'] \text{ and } \pi_R[S'] = c_R[S'].$$

We show that when $\beta > (1 - \Delta_C + \epsilon)^{1/3} + \eta$, the failure probability is less than $\frac{1}{\epsilon\eta|\mathbb{F}|}$.

Using this result, we prove that BaseFold [25] is secure in the list-decoding regime, meaning that a prover can use BaseFold to commit to a polynomial-size list of polynomials, which requires

²There exists a generic transformation for univariate to multilinear PCS, but this incurs further overhead (e.g. [19])

³See [11, 25] for more details on this comparison

⁴Actually, recent concurrent work [17] also proves Basefold and WHIR [whir] for RS codes secure in the list decoding regime, and we discuss this in detail in Section 1.2.

fewer queries from the IOPP verifier. Ultimately, this reduces the communication complexity of any SNARK that uses BaseFold as its PCS.

Technical Overview We now give a high level overview of the proof. We would like to bound the size of a set $A \subset \mathbb{F}$ of “bad” elements, where for all $r \in A$, there exists large sets $S \subset [n]$, and codewords $c_L, c_R \in C$, where

$$\pi_L[S] + r\pi_R[S] = c_L[S] + rc_R[S], \quad (1)$$

but there is no set $S' \subset S$ such that

$$\pi_L[S'] = c_L[S'], \pi_R[S'] = c_R[S'], \quad (2)$$

where $\pi_L[S]$ denotes the set $\{\pi_L[i] : i \in S\}$ (resp. for π_R) and $C[S] = \{c[S] : c \in C\}$ is the puncturing of C on the set S . First, suppose that Equation 2 holds for some set $S \subset [n]$. By linearity of C , this implies that there is only one $r \in \mathbb{F}$ such that $\pi_L[S] + r\pi_R[S] \in C[S]$. This already gives a naive upper bound on the size A , as it is *at most* the number of subsets of $[n]$ that are larger than βn , which can be computed by taking the sum of binomial coefficients (and which is bounded by 2^n). However, this is clearly insufficient as we would like $|A|/|\mathbb{F}|$ to be negligible.

Our strategy is as follows. Denote by $\mathcal{S}_{\pi, \epsilon, \eta}$ the set of “bad sets”, $S \subset [n]$ which satisfy Equations 1, 2 with respect to some codeword pair $(c_L, c_R) \in C \times C$ and some field element $r \in \mathbb{F}$. As mentioned in the previous paragraph, for any $S \in \mathcal{S}_{\pi, \epsilon, \eta}$, there is at most one $r \in \mathbb{F}$ satisfying Equations 1, 2. Using this fact, we show that $|A| \leq |\mathcal{S}_{\pi, \epsilon, \eta}|$ and going forward, we focus our attention on bounding the size of $\mathcal{S}_{\pi, \epsilon, \eta}$. Next, we observe that if $S_1, S_2 \in \mathcal{S}_{\pi, \epsilon, \eta}$ and $S_1 \cap S_2 > (1 - \Delta_C)$ then there exists some $c_L, c_R \in C \times C$ such that

$$\pi_L[S_1] + r_1\pi_R[S_1] = c_L[S_1] + r_1c_R[S_1],$$

$$\pi_L[S_2] + r_2\pi_R[S_2] = c_L[S_2] + r_2c_R[S_2],$$

for some $r_1, r_2 \in \mathbb{F}$. Furthermore, we will show that any set $S_3 \in \mathcal{S}_{\pi, \epsilon, \eta}$ such that $|S_1 \cap S_2 \cap S_3| > (1 - \Delta_C)$ also satisfies

$$\pi_L[S_3] + r_3\pi_R[S_3] = c_L[S_3] + r_3c_R[S_3],$$

for some $r_3 \in \mathbb{F}$. But on the other hand, we will show that there can’t be more than $\frac{1}{\eta}$ sets that satisfy Equations 1, 2 according to c_L, c_R . Therefore for every pair of sets, $S_1, S_2 \in \mathcal{S}_{\pi, \epsilon, \eta}$, there are at most $\frac{1}{\eta}$ sets that intersect with $S_1 \cap S_2$. Since there are at most $|\mathcal{S}_{\pi, \epsilon, \eta}|^2$ pairs of sets in $\mathcal{S}_{\pi, \epsilon, \eta}$, there are at most $|\mathcal{S}_{\pi, \epsilon, \eta}|^2 \frac{1}{\eta}$ triples of sets in $\mathcal{S}_{\pi, \epsilon, \eta}$ that have large three-way intersection. Finally we show that because $\mathcal{S}_{\pi, \epsilon, \eta}$ has only a relatively small number of triples with large three-way intersection, $\mathcal{S}_{\pi, \epsilon, \eta}$ must be smaller than $\frac{1}{\eta\epsilon}$.

BaseFold Soundness. Next, we describe how to prove soundness of BaseFold in the list decoding regime. For ease of exposition, we will describe a simplified version of the protocol where the prover sends full messages to the verifier (whereas usually the prover instead sends oracles to messages, instantiated with Merkle trees). Additionally, we only consider a one-round version of the protocol. First, the prover sends a vector $\mathbf{v}$ and the verifier responds with a random field element $r \leftarrow \mathbb{F}$. Next, the prover decomposes $\mathbf{v}$ into two vectors $\mathbf{v}_L, \mathbf{v}_R$ and sends back $\mathbf{w} = \mathbf{v}_L + r\mathbf{v}_R$. The verifier will do a probabilistic check that there exists some large set $S \subset [n]$ and codeword $c^* \in C$ such that $\mathbf{w}[S] = c^*[S]$. We want to show that this gives a high probability guarantee that $\mathbf{v}_L[S'] = c_L[S'], \mathbf{v}_R[S'] = c_R[S']$, for a large subset $S' \subset S$ and codewords $c_L, c_R \in C$ such that $c_L + rc_R = c^*$.

We will show that the following two events happen with negligible probability. In the first event, a malicious prover commits to $\mathbf{v}$ such that $\mathbf{v}_L, \mathbf{v}_R$ are far from any codeword, but then the

prover gets lucky and receives verifier challenge r such that $\mathbf{w} = \mathbf{v}_L + r\mathbf{v}_R$ is close to a codeword. The result from DeepFRI [8] already shows that this happens with negligible probability. In the second event, a malicious prover sends $\mathbf{v}_L, \mathbf{v}_R$ which agree with c_L, c_R on some large set S , but after receiving the verifier challenge, r , manages to send $\mathbf{w}$ which is close to some codeword (unlike the previous case) but does not agree with $c_L + rc_R$ on S (and is therefore possibly far from $c_L + rc_R$ while being close to some other codeword). The result presented in this paper shows that this event also only happens with negligible probability.

1.2 Related Work

There are a few concurrent and independent works that address this same problem. In “BaseFold In The List Decoding Regime” [17], Haböck proves a similar statement, but where the lower bound of β is $\sqrt{1 - \Delta_C + \epsilon}$ and where $|A| \in O(n^2)$, where $n \in \mathbb{N}$ is the number of elements in the codeword. Furthermore, their result only applies to Reed-Solomon codes. In other concurrent work, entitled “Linear Proximity Gap for Linear Codes within the 1.5 Johnson Bound” [14], they prove the statement also for general linear code, but where $|A| \in O(\frac{n}{\epsilon|\mathbb{F}|})$ when $\beta > (1 - \Delta_C + \epsilon)^{1/3}$. In Deep-Fold [16], Guo et al adapts Deep-FRI [8] to Basefold, and in that setting also proves the bound of $\beta > \sqrt{1 - \Delta_C}$, but this only applies to Reed-Solomon codes and only to the DeepFRI [8] Protocol, which is a variant of FRI with slightly more overhead. Furthermore, WHIR [2] presents an improved result in the unique decoding regime by proving a similar definition, which they call, “mutual correlated agreement” for β with lower bound $(1 - \Delta_C/2)$, whereas the previous best result in this regime ([25]) was $\beta > (1 - \Delta_C/3)$.

Additionally, there are several papers [3, 8, 9] that analyze the communication complexity of FRI [6] using the weaker version of this statement which is useful in the univariate setting. In “Worst Case To Average Case Reduction For Distance to a Linear Code” [3], the authors improve upon the original FRI paper by proving that $|A|$ is small when $\beta \geq (1 - \Delta_C)^{1/4}$ and Deep-FRI [8] improves this to $\beta \geq (1 - \Delta_C + \epsilon)^{1/3}$. Deep-FRI additionally introduces a modification of the FRI protocol which further reduces the number of verifier queries, at the cost of some (slight) prover overhead. Finally, in “Proximity Gaps of Reed-Solomon Codes” [9], the authors use a list-decoding algorithm for Reed-Solomon codes to prove correlated agreement for $\beta \geq \sqrt{1 - \Delta_C + \epsilon}$, but (as mentioned earlier with regards to [17]), this result is only meaningful with a field that is at least quadratic in the instance size and with a suitably small rate, both of which impact prover time.

2 Preliminaries

2.1 Notation

Sets Let $n \in \mathbb{Z}$. Denote by $[n]$ the set $[0, n - 1]$. For a set S , $\text{even}(S)$ is the set of even integers in S , and $\text{odd}(S)$ is the set of odd integers in S . Let $q \in \mathbb{N}$. Then $S/q = \{s/q : s \in S\}$, $S + q = \{s + q : s \in S\}$, etc. 2^S is the power set of S .

Strings and Functions Let $x \in \mathbb{F}$, $r \in \mathbb{N}$, then $x^{\parallel r}$ is the string obtained by concatenating x to itself r times. Let $f : S \rightarrow S$ be a function and $n \in \mathbb{N}$. Then $f^{\circ n}$ denotes function composition of f with itself n times.

Error-Correcting Codes We will use C to denote a linear $[n, k, d]$ code, which is a subspace C of $\mathbb{F}^n$ with an encoding algorithm $\text{Enc}_C : \mathbb{F}^k \rightarrow C$ (Definition 1). Δ_C is the minimum relative distance of the code C . Let $n \in \mathbb{N}$ and $S \subset [n]$. For a vector $\mathbf{x} \in \mathbb{F}^n$, $\mathbf{x}[S] = \{x[i] : i \in S\}$. Let $\mathbf{v} \in \mathbb{F}^n$, let C be a linear code, and let $S \subset [n]$. Then we say that $\mathbf{v}[S] \in C[S]$ if there exists a codeword $c \in C$ such that $\mathbf{v}[S] = c[S]$.

2.2 Definitions

We present a standard definition of a *linear error-correcting code*.

Definition 1 (Linear Code). *A linear error-correcting code with message length k and codeword length n is an injective mapping from $\mathbb{F}^k$ to a linear subspace $C \subseteq \mathbb{F}^n$. C is associated with a generator matrix, $G \in \mathbb{F}^{k \times n}$ such that the rows of G are a basis of C and the encoding of a vector $\mathbf{v} \in \mathbb{F}^k$ is $\mathbf{v} \cdot G$. The minimum Hamming distance of a code is the minimum on the number of different entries between any two different codewords $c_1, c_2 \in C$. If C has a minimum distance $d \in [n]$, we say that C is an $[n, k, d]$ code and use Δ_C to denote d/n —the relative minimum distance.*

3 Distance Preservation on Sets Within 1.5 Johnson Bound

In this section, we state and prove our main result. For a fixed $\pi_L, \pi_R \in \mathbb{F}^n$, we define a set of “bad” elements $r \in \mathbb{F}$, such that $\pi_L + r\pi_R$ agrees with a codeword on a “bad” set $S \subset [n]$, even when π_L, π_R do not.

Definition 2 ($\mathcal{A}_{\pi, \epsilon, \eta}$). *Let $\epsilon, \eta, \beta \in [0, 1], \beta > (1 - \Delta_C + \epsilon)^{1/3} + \eta$, $\epsilon < 0.18$, $\eta + \beta < 1$, and $\pi = (\pi_L, \pi_R) \in \mathbb{F}^{2n}$. For every $r \in \mathcal{A}_{\pi, \epsilon, \eta}$, there exists a set $S \subset [n], |S| > \beta n$, such that*

$$\pi_L[S] + r\pi_R[S] \in C[S],$$

and such that for all $S' \subset S, |S'| > (\beta - \eta)n$,

$$\pi_L[S'], \pi_R[S'] \notin C[S'].$$

Similarly, we define a set of “bad sets”.

Definition 3 ($\mathcal{S}_{\pi, \epsilon, \eta}$). *Let $r \in \mathbb{F}$, $\epsilon, \eta \in [0, 1]$ with $\beta > (1 - \Delta_C + \epsilon)^{1/3} + \eta$, $\epsilon < 0.18$, $\eta + \beta < 1$, and $\pi = (\pi_L, \pi_R) \in \mathbb{F}^{2n}$. $\mathcal{S}_{\pi, \epsilon, \eta} \subset 2^{[n]}$ is a set satisfying the following. For $r \in \mathbb{F}$, $S_r \in \mathcal{S}_{\pi, \epsilon, \eta}$ implies that $|S_r| > \beta n$ and S_r is the largest set satisfying*

$$\pi_L[S] + r\pi_R[S] \in C[S],$$

but for all $S' \subset S, |S'| > (\beta - \eta)n$,

$$\pi_L[S'], \pi_R[S'] \notin C[S'].$$

Theorem 1 (Distance Preservation on Sets Within 1.5 Johnson Bound). *Let C be a linear error-correcting code with $n \in \mathbb{N}$. Let $\epsilon, \eta \in [0, 1]$. Then for all $\pi = (\pi_L, \pi_R) \in \mathbb{F}^{2n}$,*

$$|\mathcal{A}_{\pi, \epsilon, \eta}| \leq \frac{1}{\epsilon\eta} \tag{3}$$

Throughout this proof we will introduce helpful claims as needed. For ease of exposition, we defer these smaller proofs to the end of the main proof.

Proof. We show first that $|A_\pi| \leq |\mathcal{S}_{\pi, \epsilon, \eta}|$. Looking ahead, this will enable us to focus our analysis on the size of $\mathcal{S}_{\pi, \epsilon, \eta}$. This inequality follows from the following Lemma.

Lemma 1. *There exists an injective function $f : \mathcal{A}_{\pi, \epsilon, \eta} \rightarrow \mathcal{S}_{\pi, \epsilon, \eta}$.*

Proof. Deferred to the end. □

Next, we prove the following fact about $\mathcal{S}_{A_\pi, \epsilon, \eta}$, which will enable us to bound the number of triples $S_1, S_2, S_3 \in \mathcal{S}_{A_\pi, \epsilon, \eta}$ such that $|S_1 \cap S_2 \cap S_3| > (1 - \Delta_C)$.

Lemma 2. Let $S_1, S_2 \in \mathcal{S}_{\pi, \epsilon, \eta}$ such that $|S_{r_1} \cap S_{r_2}| > (1 - \Delta_C)n$. Then there are at most $\frac{1}{\eta}$ sets $S_3 \in \mathcal{S}_{\pi, \epsilon, \eta}$ such that $|S_1 \cap S_2 \cap S_3| \geq (1 - \Delta_C)n$.

Proof. Deferred to the end. $\square$

Since there are at most $|\mathcal{S}_{\pi, \epsilon, \eta}^2|$ total pairs of sets in $\mathcal{S}_{\pi, \epsilon, \eta}$, it follows by Lemma 2 that the total number of triples, $S_1, S_2, S_3 \in \mathcal{S}_{\pi, \epsilon, \eta}$ such that $|S_1 \cap S_2 \cap S_3| \geq (1 - \Delta_C)n$ is less than or equal to

$$|\mathcal{S}_{\pi, \epsilon, \eta}|^2 \cdot \frac{1}{\eta}.$$

Next, we can use the following lemma to bound the size of $\mathcal{S}_{\pi, \epsilon, \eta}$. The proof uses a generalization of the Johnson Bound⁵

Lemma 3. Let $\epsilon, \eta, \beta \in [0, 1], n \in \mathbb{Z}$ such that $\epsilon < 0.18, \beta > (1 - \Delta_C + \epsilon)^{1/3} + \eta$, and $\eta + \beta < 1$. Then,

$$|\mathcal{S}_{\pi, \epsilon, \eta}| \leq \frac{1}{\epsilon \eta}.$$

Proof. Proof deferred. $\square$

By Lemma 1, it follows that $|\mathcal{A}_{\pi, \epsilon, \eta}| \leq |\mathcal{S}_{\pi, \epsilon, \eta}|$. Therefore,

$$|\mathcal{A}_{\pi, \epsilon, \eta}| \leq |\mathcal{S}_{\pi, \epsilon, \eta}| \leq \frac{1}{\epsilon \eta}.$$

Next, we present the deferred proofs.

Proof of Lemma 1. First define the function $f : \mathcal{A}_{\pi, \epsilon, \eta} \rightarrow \mathcal{S}_{\pi, \epsilon, \eta}$ as follows. $f(r) = S$ if $S \in \mathcal{S}_{\pi, \epsilon, \eta}$ and

$$\pi_L[S] + r\pi_R[S] \in C[S]. \quad (4)$$

First, we show that f is well defined. Suppose otherwise. Then there exists $S_1, S_2 \in \mathcal{S}_{\pi, \epsilon, \eta}$ such that $f(r) = S_1$ and $f(r) = S_2$. By definition of $\mathcal{S}_{\pi, \epsilon, \eta}$, this implies that both S_1 and S_2 are the largest sets satisfying Equation 4 with respect to r . But then $S_1 \cup S_2$ is also in $\mathcal{S}_{\pi, \epsilon, \eta}$ and satisfies that same equation. Since we assume $S_1 \neq S_2$, S_1 and S_2 are both smaller than $|S_1 \cup S_2|$, which is a contradiction. In the following claim, we show that for any set $S \in \mathcal{S}_{\pi, \epsilon, \eta}$, there is at most one $r \in \mathcal{A}_{\pi, \epsilon, \eta}$ satisfying Equation 4. Looking ahead, this will imply that f is an injection.

Claim 1. Let $\pi_L, \pi_R \in \mathbb{F}^n$. Let $S \subset [n]$. Either $\pi_L[S], \pi_R[S] \in C[S]$ or there is exactly one $r \in \mathbb{F}$ such that

$$\pi_L[S] + r\pi_R[S] \in C[S]$$

Proof of Claim 1. Suppose otherwise. Then there exists $r_1, r_2 \in \mathbb{F}$ and $c_1, c_2 \in C$ such that

$$\pi_L[S] + r_1\pi_R[S] = c_1[S],$$

$$\pi_L[S] + r_2\pi_R[S] = c_2[S],$$

and such that $\pi_L[S], \pi_R[S] \notin C[S]$. By linearity of the code, we subtract the bottom equation from the top equation and obtain

$$(r_1 - r_2) \cdot (\pi_R[S]) = (c_1 - c_2)[S].$$

⁵<https://www.cs.cmu.edu/~venkatg/teaching/au18-coding-theory/lec-scribes/list-decoding.pdf>

Again by linearity of the code, we divide both sides by $(r_1 - r_2)$, which proves that

$$\pi_R[S] = \frac{(c_1 - c_2)}{r_1 - r_2}[S].$$

Therefore, $\pi_R[S] \in C[S]$. Furthermore, plugging this back into Equation 1, we find that $\pi_L[S] \in C[S]$, which leads to a contradiction and completes the proof. $\square$

Therefore, if $f(r_1) = f(r_2)$ then $r_1 = r_2$, and therefore f is an injection. $\square$

Proof of Lemma 2. Let $S_1, S_2 \in \mathcal{S}_{\pi, \epsilon, \eta}$. Then by definition of $\mathcal{S}_{\pi, \epsilon, \eta}$, there exists $c_1, c_2 \in C$ such that

$$\pi_L[S_1] + r_1 \pi_R[S_1] = c_1[S_1], \pi_L[S_2] + r_2 \pi_R[S_2] = c_2[S_2]. \quad (5)$$

Let $S_I = S_1 \cap S_2$. By assumption of the lemma, $|S_I| > (1 - \Delta_C)n$. Since $S_I \subset S_1, S_2$, it follows that,

$$\pi_L[S_I] + r_1 \pi_R[S_I] = c_1[S_I],$$

and

$$\pi_L[S_I] + r_2 \pi_R[S_I] = c_2[S_I].$$

Since S_1, S_2 are the largest sets satisfying the Equation 5 for any given r , it follows that $r_1 \neq r_2$. By Claim 1, this implies that there exists $c_L, c_R \in C$ such that $\pi_L[S_I] = c_L[S_I]$, $\pi_R[S_I] = c_R[S_I]$. It therefore follows that for $i \in \{1, 2\}$

$$\pi_L[S_I] + r_i \pi_R[S_I] = c_i[S_I] = c_L[S_I] + r_i c_R[S_I].$$

Since $|S_I| > (1 - \Delta_C)n$, this implies that the equality holds over the entire set S_1 (resp S_2), because otherwise two codewords agree on more than $(1 - \Delta_C)n$ positions which is a contradiction since Δ_C is the minimum distance of the code and therefore $(1 - \Delta_C)n$ is the maximum agreement between any two distinct codewords. Therefore for $i \in \{1, 2\}$,

$$\pi_L[S_i] + r_i \pi_R[S_i] = c_L[S_i] + r_i c_R[S_i]. \quad (6)$$

Next, we show that if a third set $S_3 \in \mathcal{S}_{\pi, \epsilon, \eta}$ has large intersection with S_1, S_2 then Equation 6 is also true for $i = 3$.

Claim 2. Let $S_3 \in \mathcal{S}_{\pi, \epsilon, \eta}$, and let $S'_I = S_1 \cap S_2 \cap S_3$ such that $|S'_I| > (1 - \Delta_C)n$. Then there exists $r_3 \in \mathbb{F}$ such that

$$\pi_L[S_3] + r_3 \pi_R[S_3] = c_L[S_3] + r_3 c_R[S_3].$$

Proof. $S'_I \subset S_I$ and so

$$\pi_L[S'_I] + r_3 \pi_R[S'_I] = c_3[S'_I] = c_L[S'_I] + r_3 c_R[S'_I].$$

Since $|S'_I| \geq (1 - \Delta_C)n$, this implies that

$$\pi_L[S_3] + r_3 \pi_R[S_3] = c_L[S_3] + r_3 c_R[S_3].$$

$\square$

Finally we prove that the number of sets S_i that satisfy Equation 6 is less than $\frac{1}{\eta}$.

Claim 3. Let $c_L, c_R \in C$ and define $\mathcal{S}_{(c_L, c_R)} \subseteq \mathcal{S}_{\pi, \epsilon, \eta}$ to be the subset such that for each $S \in \mathcal{S}_{(c_L, c_R)}$, there exists $r \in \mathcal{A}_{\pi, \epsilon, \eta}$ such that

$$\pi_L[S] + r \pi_R[S] = c_L[S] + r c_R[S].$$

Then

$$|\mathcal{S}_{(c_L, c_R)}| \leq \frac{1}{\eta}.$$

Proof. For $S \in \mathcal{S}_{(c_L, c_R)}$, define $na(S)$ as follows.

$$na(S) = \{i \in S : c_L[i] \neq \pi_L[i] \wedge c_R[i] \neq \pi_R[i]\}.$$

Let $S_1, S_2 \in \mathcal{S}_{(c_L, c_R)}$. By definition of $\mathcal{S}_{\pi, \epsilon, \eta}$, it follows that there exists $r_1, r_2 \in \mathbb{F}$, with $r_1 \neq r_2$ such that

$$\pi_L[na(S_1)] + r_1 \pi_R[na(S_1)] = c_L[na(S_1)] + r_1 c_R[na(S_1)].$$

$$\pi_L[na(S_2)] + r_2 \pi_R[na(S_2)] = c_L[na(S_2)] + r_2 c_R[na(S_2)].$$

Therefore, since $\pi_L[na(S_1)] \notin C[na(S_1)]$, it follows from Claim 1 that

$$na(S_1) \cap na(S_2) = \emptyset,$$

Therefore

$$\left| \bigcup_{S \in \mathcal{S}_{(c_L, c_R)}} na(S) \right| = \sum_{S \in \mathcal{S}_{(c_L, c_R)}} |na(S)| \leq n.$$

Furthermore, by definition of $\mathcal{S}_{\pi, \epsilon, \eta}$, for each $S \in \mathcal{S}_{(c_L, c_R)}$, $|na(S)| \geq \eta n$. Therefore,

$$|\mathcal{S}_{(c_L, c_R)}| \cdot \eta n \leq \sum_{S \in \mathcal{S}_{(c_L, c_R)}} |na(S)|,$$

and so, combining the previous two equations,

$$|\mathcal{S}_{(c_L, c_R)}| \leq \frac{\sum_{S \in \mathcal{S}_{(c_L, c_R)}} |na(S)|}{\eta n} \leq \frac{n}{n\eta} = \frac{1}{\eta}$$

□

□

Proof of Lemma 3. We construct a bipartite graph, G , where right vertices are labeled by elements in $[n]$ and left vertices are labeled by sets in $\mathcal{S}_{\pi, \epsilon, \eta}$.

This proof is a generalization of the proof of the Johnson bound ⁶. For $v_0, v_1, v_2 \in L$ and $w \in R$, define a “3-angle” as an 4-tuple, v_0, v_1, v_2, w , such that $(v_i, w) \in E$ for all $i \in [3]$. Let $d(w)$ be the degree of node w . Then the number of 4-angles in the graph is equal to

$$\sum_{v \in R} \binom{d(v)}{3}$$

By Jensen’s Inequality, we know that,

$$\sum_{v \in R} \binom{d(v)}{3} \geq |R| \binom{(\sum_{v \in R} d(v))/|R|}{3}. \quad (7)$$

Since each set in $\mathcal{S}_{\pi, \epsilon, \eta}$ has greater than or equal to βn elements, it follows that each vertex in L has more than βn neighbors. Furthermore, the sum of the degrees of all vertices in R is greater than $|L| \cdot \beta |R|$, i.e. $\sum_{v \in R} d(v) \geq |L| \cdot \beta |R|$. Thus, by Equation 7, it follows that,

$$\sum_{v \in R} \binom{d(v)}{3} \geq n \binom{\beta |L|}{3}.$$

⁶<https://www.cs.cmu.edu/~venkatg/teaching/au18-coding-theory/lec-scribes/list-decoding.pdf>

Next, we prove an upper bound on the number of 3-angles in G . By assumption of the lemma, there are at most $|\mathcal{S}_{\pi,\epsilon,\eta}|^2 \frac{1}{\eta}$ triples, $S_1, S_2, S_3 \in \mathcal{S}_{\pi,\epsilon,\eta}$ such that $|S_1 \cap S_2 \cap S_3| > (1 - \Delta_C)n$. Furthermore, $|S_1 \cap S_2 \cap S_3| \leq \beta n$ because otherwise by Claim 1, $\pi_L[S_1] \in C[S_1]$ and $|S_1| > \beta > \beta - \eta$, which contradicts the definition of $\mathcal{S}_{\pi,\epsilon,\eta}$. All other triples in $\mathcal{S}_{\pi,\epsilon,\eta}$ have three-way intersection less than or equal to $(1 - \Delta_C)n < (\beta - \eta)^3 - \epsilon \leq \beta^3 - \epsilon$. Translating this to the graph G , we find that the total number of angles in G is at most

$$\binom{|L|}{3}(\beta^3 - \epsilon)n + |L|^2 \frac{1}{\eta} \beta n.$$

Combining the two inequalities, we have,

$$\binom{\beta|L|}{3} \leq \binom{|L|}{3}(\beta^3 - \epsilon) + |L|^2 \frac{1}{\eta} \beta.$$

Let $\alpha := \beta^3 - \epsilon$. Then,

$$\frac{\beta|L|(\beta|L| - 1)(\beta|L| - 2)}{3} \leq \frac{\alpha|L|(|L| - 1)(|L| - 2)}{3} + |L|^2 \frac{1}{\eta} \beta$$

$$\beta|L|(\beta|L| - 1)(\beta|L| - 2) \leq |L|[\alpha(|L| - 1)(|L| - 2) + 3|L|\frac{1}{\eta}\beta]$$

Factor out L , multiply both sides by 3

$$\beta(\beta|L| - 1)(\beta|L| - 2) \leq \alpha(|L| - 1)(|L| - 2) + 3|L|\frac{1}{\eta}\beta$$

Divide both sides by L

$$(\beta^2|L| - \beta)(\beta|L| - 2) \leq (\alpha|L| - \alpha)(|L| - 2) + 3|L|\frac{1}{\eta}\beta$$

Multiply first two terms

$$(\beta^3|L|^2 - 2\beta^2|L| - \beta^2|L| + 2\beta) \leq \alpha|L|^2 - 2\alpha|L| - \alpha|L| + 2\alpha + 3|L|\frac{1}{\eta}\beta$$

Expand terms

$$(\beta^3|L|^2 - 3\beta^2|L| + 2\beta) \leq \alpha|L|^2 - 3|L|(\alpha - \frac{1}{\eta}\beta) + 2\alpha$$

Combine terms

$$(\beta^3 - \alpha)|L|^2 - 3|L|(\beta^2 - \alpha + \frac{1}{\eta}\beta) + 2(\beta - \alpha) \leq 0$$

Subtract RHS from both sides

Let $P(X) = (\beta^3 - \alpha)X^2 - 3(\beta^2 - \alpha + \frac{1}{\eta})X + 2(\beta - \alpha)$. Our goal is to solve for L , such that $P(L) \leq 0$. Since P is concave, it is smaller than 0 for X between $[x_0, x_1]$, where x_0, x_1 are its two roots and $x_0 < x_1$. Next, we solve for x_1 , which is an upper bound on L . By the quadratic equation,

$$x_1 = \frac{3(\beta^2 - \alpha) + \sqrt{(-3(\beta^2 - \alpha))^2 - 8(\epsilon)(\beta - \alpha)}}{2(\epsilon)}$$

Next, we use the following claim to show that x_1 is a real root.

Claim 4.

$$(-3(\beta^2 - (\beta^3 - \epsilon)))^2 - 8\epsilon(\beta - (\beta^3 - \epsilon)) > 0.$$

Proof. We will show that

$$(-3(\beta^2 - (\beta^3 - \epsilon)))^2 > 8\epsilon(\beta - (\beta^3 - \epsilon)).$$

Equivalently, we show that,

$$9(\beta^2 - (\beta^3 - \epsilon))^2 > 8\epsilon(\beta - (\beta^3 - \epsilon)).$$

Define the function $f_\beta(\epsilon)$ as follows.

$$f_\beta(\epsilon) = 9(\beta^2 - (\beta^3 - \epsilon))^2 - 8\epsilon(\beta - (\beta^3 - \epsilon)) = \epsilon^2 + (18(\beta^2 - \beta^3) - 8(\beta - \beta^3))\epsilon + 9(\beta^2 - \beta^3)^2.$$

Since $\beta > (1 - \Delta_C + \epsilon)^{1/3} + \eta$, it follows that $\epsilon < \beta^3$, because otherwise $(\beta - \eta)^3 \geq (1 - \Delta_C + \beta^3)$, which implies that $\beta^3 > (1 - \Delta_C + \beta^3)$, which implies that $0 > (1 - \Delta_C)$, which is a contradiction. We show that for all $\beta \in [0, 1]$, and for all $\epsilon \in [0, \beta^3]$, $f_\beta(\epsilon) > 0$. First,

$$f_\beta(\beta^3) = \beta^4 > 0.$$

Next, it is easy to verify that when $\beta < 0.5$, $f_\beta(\epsilon)$ has two real roots, both of which are greater than β^3 , and if $\beta \geq 0.5$, then $f_\beta(\epsilon)$ has no real roots. In both cases, this implies that $f_\beta(\epsilon)$ is positive for $\epsilon \in [0, \beta^3]$. □

Therefore, since x_1 is a real root, we have,

$$\begin{aligned} x &\leq \frac{3(\beta^2 - \alpha + \frac{1}{\eta}\beta) + \sqrt{(-3(\beta^2 - \alpha + \frac{1}{\eta}\beta))^2}}{2(\epsilon)} \\ &= \frac{3(\beta^2 - \alpha + \frac{1}{\eta}\beta) + 3(\beta^2 - \alpha + \frac{1}{\eta}\beta)}{2\epsilon} \\ &= \frac{2 \cdot (3(\beta^2 - \alpha + \frac{1}{\eta}\beta))}{2\epsilon} = \frac{3(\beta^2 - \alpha + \frac{1}{\eta}\beta)}{\epsilon} \\ &= \frac{3(\beta^2 - (\beta^3 - \epsilon) + \frac{1}{\eta}\beta)}{\epsilon} \\ &= \frac{3(\beta^2 - (\beta^3 - \epsilon))}{\epsilon} + \frac{\frac{1}{\eta}\beta}{\epsilon} \end{aligned}$$

Next, we use the following claim to bound the numerator.

Claim 5. *Let $\epsilon \in [0, 1]$ such that $\epsilon < \beta^3$ and $\epsilon < 0.18$. Then,*

$$\beta^2 - (\beta^3 - \epsilon) < \frac{1}{3}.$$

Proof. First, the function $f(\beta) = \beta^2 - \beta^3$ has a maximum at $\beta = \frac{2}{3}$, where $f(\frac{2}{3}) \approx 0.148$. Therefore, since $\epsilon < 0.18$, it follows that

$$\begin{aligned} &\beta^2 - (\beta^3 - \epsilon) \\ &= \beta^2 - \beta^3 + \epsilon \\ &< 0.148 + 0.18 \\ &= 0.32 < 1/3 \end{aligned}$$

□

Therefore,

$$|L| \leq x_1 \leq \frac{1}{\epsilon} + \frac{\beta}{\epsilon\eta} \leq \frac{\eta + \beta}{\epsilon\eta} \leq \frac{1}{\epsilon\eta},$$

where the last inequality follows by the assumption that $\epsilon + \eta < 1$. This completes the proof. □

□

4 Improved Soundness of the BaseFold Protocol

In this section, we present a new soundness analysis for BaseFold, and show that the soundness error remains small even if the verifier only makes $l := \frac{\lambda}{\log_2(\beta)}$ queries for $\beta \geq (1 - \Delta + \epsilon)^{1/3} + \eta$. Previously, this was only proven for $\beta \geq 1 - \Delta/3 + \eta$. In this paper, we only focus on the BaseFold IOP for evaluation of a polynomial at a random point. The BaseFold paper [25] describes how to obtain evaluation at an arbitrary point from evaluation at a random point. The BaseFold IOP for evaluation at a random point remains unchanged from [25]⁷. We restate the IOP in Figure 1 for completeness. We also restate the definition of a *foldable code*⁸, which was introduced in BaseFold [25]. A *foldable code* is a family of codes, which we will denote $(C_d, \dots, C_0)$, which are characterized by a set of vectors $\{\mathbf{t}_i \in \mathbb{F}^{n_i} : i \in [1, d]\}$. Each codeword $c_i \in C_i$ is composed of two codewords in C_{i-1} . Additionally, the structure of a *foldable codes* enables local consistency checks between codewords in adjacent codes. These consistency tests allow the BaseFold⁹ IOPs to maintain logarithmic query complexity.

Definition 4 (Foldable Code). *Let $c, d \in \mathbb{N}$ and for each $i \in [d]$, define $k_i = 2^i, n_i = c \cdot k_i$. A $[n_d, k_d]$, foldable code is a family of codes $(C_0, \dots, C_d)$, where the base code, C_0 , is equal to the repetition code, $\{m^{||c} : m \in \mathbb{F}\}$ and each $[n_i, k_i]$ code, C_i , and each $\mathbf{c} \in C_i$ satisfies the following:*

$$\begin{aligned} \mathbf{c} := \text{Enc}_{C_i}(\mathbf{m}) &= \text{Enc}_{C_{i-1}}(\mathbf{m}_L) + \mathbf{t}_i \circ \text{Enc}_{C_{i-1}}(\mathbf{m}_R) \\ &|| \text{Enc}_{C_{i-1}}(\mathbf{m}_L) - \mathbf{t}_i \circ \text{Enc}_{C_{i-1}}(\mathbf{m}_R) \end{aligned}$$

where $\{\mathbf{t}_i \in \mathbb{F}^{n_i} : i \in [1, d]\}$ is given in the description of the code, $\mathbf{m} = (\mathbf{m}_L || \mathbf{m}_R)$ is a vector in $\mathbb{F}^{k_i}$, and $\circ$ denotes the Hadamard product.

Remark 1. *In the above definition, elements $\mathbf{c}[i]$ and $\mathbf{c}[i + n_i/2]$ are two points on the same line for each $i \in [n_i/2]$. For the remainder of this section, we will assume that the codeword has been re-ordered, so that $\mathbf{c}[i], \mathbf{c}[i + 1]$ are on the same line for each $i \in \text{even}([n_i])$. It is easy to prove that folding preserves this ordering.*

Remark 2. *It was proven in BaseFold [25] that foldable codes are equivalent to punctured Reed-Muller codes over a domain with nice properties. We will use this fact in the proof of soundness.*

Foldable codes are attractive because a codeword in C_i can be transformed into a smaller codeword in C_{i-1} using only local operations. More specifically, we query the same random point on each of the $n_i/2$ lines defined by $\mathbf{c}[j] + X\mathbf{c}[j + 1]$ $\{(\mathbf{c}[j], \mathbf{c}[j + 1]) : j \in \text{even}([n_i])\}$, and obtain a new codeword in C_{i-1} . We describe this formally with the following definition.

Definition 5 (Fold). *Define $\text{interp} : \mathbb{F}^2 \times \mathbb{F}^2 \rightarrow \mathbb{F}[X]$ to be Lagrange Interpolation of a degree-one univariate polynomial. Let $(C_d, \dots, C_0)$ be a family of foldable codes characterized by $\{\mathbf{t}_i \in \mathbb{F}^{n_i} : i \in [1, d]\}$. For each $\mathbf{v} \in C_i$, and $j \in \text{even}([n_i])$, define the pair (p_j, p_{j+1}) as follows.*

$$(p_j, p_{j+1}) = ((\mathbf{t}[j], \mathbf{v}[j]), (-\mathbf{t}[j], \mathbf{v}[j + 1])).$$

Then, the fold of $\mathbf{v}$ with respect to $r \in \mathbb{F}$ is the vector, $\text{fold}(\mathbf{v}, r)[j]$ satisfying,

$$\text{fold}(\mathbf{v}, r)[j] = \text{interp}(p_j, p_{j+1})(r).$$

⁷The syntax of this description is slightly different than that in [25], but the protocol itself is equivalent

⁸For ease of exposition, we define the codes according to a different ordering than the original.

⁹This structure also underlies certain types of Reed-Solomon codes and is the reason for FRI's efficiency

At times, we will need to work with the univariate polynomials defined by $\text{interp}(p_j, p_{j+1})$ directly. We call these polynomials the *unfolding* of $\mathbf{v}$. To ease exposition, we denote by $\mathbf{v}_L, \mathbf{v}_R$ the codewords that for all $j \in \text{even}([n_i])$ satisfy,

$$\text{interp}(p_j, p_{j+1}) = \mathbf{v}_L[j] + X\mathbf{v}_R[j]. \quad (8)$$

We remark that fold can also be defined over arbitrary vectors that are not codewords, and indeed the FRI and Basefold IOPPs rely on this fact. For a generic $\pi \in \mathbb{F}^{n_i}$, define the pair of points $(p_j, p_{j+1}) = ((\mathbf{t}[j], \pi[j]), (-\mathbf{t}[j], \pi[j+1]))$. Then, as before $\text{fold}(\pi, r) = \text{interp}(p_j, p_{j+1})$. Finally, we will sometimes fold over entire sets $S \subset [n_i]$, and this operation is well defined as long as S contains $j+1$ whenever it contains j . We introduce additional notation for this as follows.

$$\text{fold}(\pi, r)[S] = \{\text{interp}((\mathbf{t}_i[j], \pi[j]), (-\mathbf{t}_i, \pi[j+1]))(r) : j \in \text{even}(S)\} \quad (9)$$

Protocol 1 IOPP.commit

Input oracle: $\pi_d \in \mathbb{F}^{n_d}$

Output oracles: $(\pi_{d-1}, \dots, \pi_0) \in \mathbb{F}^{n_{d-1}} \times \dots \times \mathbb{F}^{n_0}$

- For i from $d-1$ downto 0:
 1. The verifier samples and sends $\alpha_i \leftarrow \$ \mathbb{F}$ to the prover
 2. For each index $j \in \text{even}[0, n_{i+1}-1]$, the prover
 - (a) sets $f(X) := \text{interp}((\mathbf{diag}(T_i)[j], \pi_{i+1}[j]), (\mathbf{diag}(-T_i)[j], \pi_{i+1}[j+1]))$
 - (b) sets $\pi_i[j] = f(\alpha_i)$
 3. The prover outputs oracle $\pi_i \in \mathbb{F}^{n_i}$.
-

Protocol 2 IOPP.query

Oracles: $(\pi_d, \dots, \pi_0)$

Repetition Parameter: $\lambda \in \mathbb{N}$

- For $j \in [0, \lambda-1]$
 - The verifier samples an index $\mu_j \leftarrow \$ \text{even}[1, n_d-1]$
 - For i from $d-1$ downto 0, the verifier
 1. queries oracle entries $\pi_{i+1}[\mu_j], \pi_{i+1}[\mu_j+1]$
 2. computes $p(X) := \text{interpolate}((\mathbf{diag}(T_i)[\mu_j], \pi_{i+1}[\mu_j]), (\mathbf{diag}(-T_i)[\mu_j], \pi_{i+1}[\mu_j+1]))$
 3. checks that $p(\alpha_i) = \pi_i[\mu_j/2]$
 4. if $i > 0$ and $\mu_j/2 \bmod 2 = 0$, update $\mu_j \leftarrow \mu_j/2$, otherwise update $\mu_j \leftarrow \mu_j/2 - 1$.
 - If π_0 is a valid codeword w.r.t. generator matrix C_0 , output **accept**, otherwise output **reject**
-

Figure 1: The IOPP protocol for foldable codes.

Theorem 2 (Soundness of Basefold IOP). *Let $\lambda \in \mathbb{N}$ be a security parameter, let $d \in \mathbb{N}$ be the number of rounds in the IOP, $\pi_d \in \mathbb{F}^{n_d}$, $l \in \mathbb{N}$, $\epsilon, \eta \in [0, 1]$, with $\beta > (1 - \Delta_C + \epsilon)^{1/3} + d\eta$, $\eta + \beta < 1$, $\epsilon < 0.18$, and $\beta^l \leq \text{negl}(\lambda)$. Then, with probability greater than*

$$1 - \frac{d}{\epsilon\eta|\mathbb{F}|},$$

*over verifier randomness $(r_d, \dots, r_1) \leftarrow \mathbb{F}$ in the **commit** phase, and letting $\{\pi_i \in \mathbb{F}^{n_i} : i \in [d+1]\}$ be the corresponding oracles sent by the prover, either the verifier accepts with probability less than*

$$\beta^l,$$

or $\exists P \in \mathbb{F}[X_1, \dots, X_d]$ such that $\text{Enc}_{C_0}(P(r_d, \dots, r_1)) = \pi_0$ and $\Delta(\text{Enc}_{C_d}(P), \pi_d) < (1 - (\beta - d\eta))n_d$.

High Level Overview of Proof Recall from Theorem 1, that if $\pi_L[S] + r\pi_R[S] = c[S]$ for some $c \in C$, then $\pi_L[S], \pi_R[S]$ differ from $c_L[S], c_R[S]$ in very few locations, where $c_L + rc_R = c$. We will show in Lemma 4, that when C is a foldable code (equivalently a punctured Reed-Muller code, See Appendix F in [25]), c is the encoding of polynomial P , c_L is the encoding of polynomial P_L and c_R is the encoding of polynomial P_R where $P_L + rP_R = P$. Next, in Lemma 5, we prove soundness for just one round of the IOP, and finally, we show how to extend this to the full, multi-round IOP.

Lemma 4. *Let $n, k \in \mathbb{N}$ and let C be an $[n, k]$ linear error-correcting code. Let $\beta, \tau_1, \tau_2 \in [0, 1]$ such that $\beta - (\tau_1 + \tau_2) > 1 - \Delta_C$. Let $\pi_L, \pi_R \in \mathbb{F}^n$, $d = \log_2(n)$. Suppose that $S \subset [n]$ where $|S| > \beta n$. Next, denote by S_1, S_2 the following sets.*

$$S_1(P_L, P_R) = \{i \in S : \pi_L[i] = \text{Enc}_C(P_L)[i] \wedge \pi_R[i] = \text{Enc}_C(P_R)[i]\},$$

where $P_L, P_R \in \mathbb{F}[X_1, \dots, X_d]$ and,

$$S_2(c, r) = \{i \in S : \pi_L[S] + r\pi_R[S] = c[S]\},$$

where $c \in C, r \in \mathbb{F}$. Suppose there exists $P_L, P_R \in \mathbb{F}[X_1, \dots, X_d]$ such that

$$|S_1(P_L, P_R)| > (|S| - \tau_1)n,$$

and suppose there exists $c \in C, r \in \mathbb{F}$ such that

$$|S_2(c, r)| > (|S| - \tau_2)n \tag{10}$$

Then

$$c = \text{Enc}_C(P_L + rP_R). \tag{11}$$

Proof. Let $c_L := \text{Enc}_C(P_L), c_R := \text{Enc}_C(P_R)$, $r \in \mathbb{F}$, $\pi^* := \pi_L + r\pi_R$, $P^*(X_1, \dots, X_d) = P_L + rP_R$ and $c^* = \text{Enc}_C(P^*)$. Then,

$$\begin{aligned} & |\{i \in S : \pi^*[i] = c^*[i]\}| \\ &= |\{i \in S : \pi_L[i] + r\pi_R[i] = \text{Enc}_C(P_L + rP_R)[i]\}| && \text{(By definition of } \pi^*, c^*, P^*) \\ &= |\{i \in S : \pi_L[i] + r\pi_R[i] = \text{Enc}_C(P_L)[i] + r\text{Enc}_C(P_R)[i]\}| && \text{(By linearity of } C) \\ &\geq |\{i \in S : \pi_L[i] = \text{Enc}_C(P_L)[i] \wedge \pi_R[i] = \text{Enc}_C(P_R)[i]\}| && \text{(By linearity of } C) \\ &= S_1(P_L, P_R) && \text{(By definitions of } S_1) \end{aligned} \tag{12}$$

Let $c \in C$ be the codeword satisfying Equation 10. Then,

$$\begin{aligned}
& |\{i \in S : c^*[i] = c[i]\}| \\
& > |\{i \in S : \pi^*[i] = c^*[i] \wedge \pi^*[i] = c[i]\}| \\
& = |\{i \in S : \pi^*[i] = c^*[i]\} \cap \{\pi^*[i] = c[i]\}| && \text{By definition of intersection} \\
& = |S_1(P_L, P_R) \cap S_2(c, r)| && \text{By Equation 12 and definition of } S_2 \\
& > |\S_1(P_L, P_R)| + |S_2(c)| - |S_1(P_L, P_R) \cup S_2(c)| && \text{By properties of set intersection} \\
& = |S_1(P_L, P_R)| + |S_2(c)| - |S_1(P_L, P_R) \cup S_2(c)| && \text{By Equation 12} \\
& > (|S| - \tau_1) + (|S| - \tau_2) - |S_1(P_L, P_R) \cup S_2(c)| && \text{By assumption of the lemma} \\
& > 2|S| - (\tau_1 + \tau_2) - |S| && |S_1(P_L, P_R) \cup S_2(c)| \leq |S| \\
& = |S| - (\tau_1 + \tau_2) && \text{Combine terms and simplify} \\
& > \beta - (\tau_1 + \tau_2) && \text{By assumption that } |S| > \beta
\end{aligned}$$

By assumption of the lemma, $\beta - (\tau_1 + \tau_2) > 1 - \Delta_C$. Therefore, by minimum distance properties of C , $c^* = c$, which completes the proof. $\square$

Next, we combine Lemma 4 with Theorem 1 to prove soundness over one single round of the IOP.

Lemma 5 (One Round Soundness). *Let $d \in \mathbb{N}$ and let C_d, C_{d-1} be a pair of codes from an $[n_d, k_d]$ foldable code family (Definition 4). Let $\pi \in \mathbb{F}^{n_d}, \epsilon, \eta \in [0, 1]$ and let $\beta > (1 - \Delta_C + \epsilon)^{1/3} + \eta$ be defined as in Theorem 1. Suppose that $\beta - 2d\eta > 1 - \Delta_{C_d}$, and that $\exists S \subset [n_d]$ and $c \in C_{d-1}$ such that*

$$|\{i \in S : \text{fold}(\pi, r)[i] = c[i]\}| > (\beta - (d-1)\eta)n_{d-1}. \quad (13)$$

Then with probability greater than $1 - \frac{d}{\epsilon\eta|\mathbb{F}|}$ (over verifier randomness r) there exists $P_L, P_R \in \mathbb{F}[X_1, \dots, X_{d-1}]$ such that $c = \text{Enc}_{C_{d-1}}(P_L + rP_R)$ and

$$|\{i \in S : \pi[i] = \text{Enc}_{C_d}(P_L + XP_R)[i]\}| > (\beta - d\eta)n_d. \quad (14)$$

Proof. By Equation 13 and by Definition of fold (Definition 4), it follows that

$$|\{i \in S : \pi_L[i] + r\pi_R[i] = c[i]\}| > (\beta - (d-1)\eta)n_{d-1},$$

where (π_L, π_R) is the *unfolding* (Definition 8) of π . It follows from Theorem 1 that if $r \notin \mathcal{A}_{\pi, \epsilon, \eta}$ then there exists $c_L, c_R \in C_{d-1}$ such that $c_L + rc_R = c$ and,

$$|\{i \in S : \pi_L[i] = c_L[i] \wedge \pi_R[i] = c_R[i]\}| > (\beta - (d-1)\eta - \eta)n_{d-1} = (\beta - d\eta)n_{d-1}. \quad (15)$$

Let $P_L, P_R \in \mathbb{F}[X_1, \dots, X_d]$ satisfy $c_L = \text{Enc}_{C_{d-1}}(P_L), c_R = \text{Enc}_{C_{d-1}}(P_R)$. Then by definition of a foldable code (Definition 4), it follows that

$$|\{i \in S : \pi[i] = \text{Enc}_{C_d}(P_L + XP_R)[i]\}| \geq 2(\beta - d\eta)n_{d-1} = (\beta - d\eta)n_d.$$

Furthermore, by Lemma 4,

$$c = \text{Enc}_{C_{d-1}}(P_L + rP_R).$$

By Theorem 1, $|\mathcal{A}_{\pi, \epsilon, \eta}| \leq \frac{1}{\epsilon\eta}$, and therefore the probability that $r \notin \mathcal{A}_{\pi, \epsilon, \eta}$ is greater than $1 - \frac{1}{\epsilon\eta|\mathbb{F}|}$, which completes the proof. $\square$

Next, we show that if the verifier accepts with probability greater than β^l , then this implies the existence of d large sets, one in each oracle, that are consistent with each other with respect to the fold operation.

Lemma 6 (Verifier Queries). *Let $\epsilon, \eta \in [0, 1]$, let $\beta > (1 - \Delta_C + \epsilon) + \eta$ be defined as in Theorem 1, and let $l \in \mathbb{N}$. If the verifier accepts the query phase with probability greater than β^l then there exists d large sets $\{f_i(S) \subset [n_i] : i \in [d], |f_i(S)| > \beta n_i\}$ such that for all $i \in [d]$,*

$$\text{fold}(\pi_{i+1}, \mathbf{r}[i+1])[f_{i+1}(S)] = \pi_i[\text{even}(f_{i+1}(S))/2] \quad (16)$$

where fold is defined in Equation 9.

Proof. Define the function $Q : \text{even}([n_d]) \rightarrow \{0, 1\}$ as $Q(\mu) = 1$ if the unique verifier query beginning with $\mu \leftarrow \text{even}(n_d)$ (defined in Protocol 2) passes the verifier tests and $Q(\mu) = 0$ otherwise. Let $S = Q^{-1}(1)$. Then, each verifier sample is a Bernoulli trial with success probability $\frac{|S|}{|\text{even}([n_d])|}$. After l queries, the probability of acceptance is $(\frac{|S|}{|\text{even}([n_d])|})^l$. Therefore, if the verifier accepts with probability greater than β^l , then $(\frac{|S|}{|\text{even}([n_d])|})$ must be larger than β , and so $|S| > \beta |\text{even}([n_d])| = \beta n_{d-1}$. Next, we define $f_i(S)$.

Definition 6 ($f_d(S)$). *Let $d \in \mathbb{N}$ and $S \subset \text{even}([n_d])$. Then,*

$$f_d(S) = S \cup (S + 1).$$

For $i \in [d]$, $f_i(S)$ satisfies the following:

$$\text{even}(f_i(S)) = \{\text{even}(\{j/2, j/2 - 1\}) : j \in \text{even}(f_{i+1}(S))\}$$

$$\text{odd}(f_i(S)) = \text{even}(f_i(S)) + 1.$$

To complete the Lemma, we need to prove that for each $i \in [d + 1]$,

$$\text{fold}(\pi_i, \mathbf{r}[i])[f_i(S)] = \pi_{i-1}[\text{even}(f_i(S))/2].$$

For each $\mu \in S$, let $((\mu_d, \mu_d + 1), \dots, (\mu_1, \mu_1 + 1))$ be the unique set of queries associated with μ (defined in Protocol 2). Then, by definition of fold (Definition 9), for each $i \in [1, d]$

$$\text{fold}(\pi_i, r_i)[f_i(S)] = \{\pi_{i,L}[\mu_i] + r_i \pi_{i,R}[\mu_i + 1]\} \quad (17)$$

Furthermore,

$$\pi_{i,L}[\mu_i] + r_i \pi_{i,R}[\mu_i + 1] = \pi_{i-1}[\mu_i/2]. \quad (18)$$

Combining Equations 17 and 18 gives

$$\text{fold}(\pi_i, r_i)[f_i(S)] = \{\pi_{i-1}[\mu_i/2] : \mu_i \in \text{even}(f_i(S))\} = \pi_{i-1}[\text{even}(f_i(S))/2],$$

which completes the proof. $\square$

Finally, we are ready to prove the main statement of the Theorem.

Proof. Suppose by contradiction that the verifier accepts with probability $> \beta^l$ but there does not exist $P \in \mathbb{F}[X_1, \dots, X_d]$ such that $\text{Enc}_{C_0}(P(r_d, \dots, r_1)) = \pi_0$ and $\Delta(\text{Enc}_{C_d}(P), \pi_d) < (1 - (\beta - d\eta))$. By Lemma 6, there exists a set $S \subset \text{even}(n_d)$ such that for each $i \in [1, d]$,

$$\text{fold}(\pi_i, \mathbf{r}[i])[f_i(S)] = \pi_{i-1}[\text{even}(f_i(S))/2].$$

Therefore, since $\pi_0 \in C_0$ there must exist a round where Equation 13 holds but Equation 14 does not. By Lemma 5, this only happens with probability $\frac{1}{\epsilon\eta|\mathbb{F}|}$. Taking the union bound over d rounds completes the proof. $\square$



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